Day **25 – Reverse Linked List**

🚀 Reverse Linked List

💡 Problem

Given a linked list:

1 → 2 → 3 → 4 → 5 → NULL

👉 Output hona chahiye:

5 → 4 → 3 → 2 → 1 → NULL

🧠 Intuition (Sabse Important)

👉 Normally list left → right jaati hai
👉 Hume direction reverse karna hai

Matlab:

1 → 2 ko 1 ← 2 banana hai

Har node ka next pointer ulta karna hai

🔑 3 Pointer Approach (MOST IMPORTANT)

Hum 3 pointer use karte hain:

prev = null
curr = head
next = null

🔄 Steps

Har step me:

1. next = curr.next
2. curr.next = prev
3. prev = curr
4. curr = next

🧪 Dry Run (Step by Step)

Initial:

prev = NULL
curr = 1 → 2 → 3 → 4 → 5

Iteration 1:

next = 2
1 → NULL
prev = 1
curr = 2

Iteration 2:

next = 3
2 → 1 → NULL
prev = 2
curr = 3

Iteration 3:

3 → 2 → 1 → NULL

Final:

5 → 4 → 3 → 2 → 1 → NULL

👉 At end:

head = prev

✅ Java Code (Clean & Interview Ready)

```java
package Day_25;

public class ReverseLinkedList {

    class Node {
        int data;
        Node next;

        Node(int data) {
            this.data = data;
            this.next = null;
        }
    }

    Node head;

    // Reverse Linked List
    public void reverseList() {
        Node prev = null;
        Node curr = head;
        Node next = null;

        while (curr != null) {
            next = curr.next;   // step 1
            curr.next = prev;   // step 2
            prev = curr;        // step 3
            curr = next;        // step 4
        }

        head = prev;
    }

    // Print List
    public void printList() {
        Node temp = head;
        while (temp != null) {
            System.out.print(temp.data + " -> ");
            temp = temp.next;
        }
        System.out.println("NULL");
    }

    // Insert at end (helper)
    public void insert(int data) {
        Node newNode = new Node(data);

        if (head == null) {
            head = newNode;
            return;
        }

        Node temp = head;
        while (temp.next != null) {
            temp = temp.next;
        }

        temp.next = newNode;
    }

    public static void main(String[] args) {
        ReverseLinkedList list = new ReverseLinkedList();

        list.insert(1);
        list.insert(2);
        list.insert(3);
        list.insert(4);
        list.insert(5);
```

```java

        System.out.println("Original List:");
        list.printList();

        list.reverseList();

        System.out.println("Reversed List:");
        list.printList();
    }
}

```

⏱️ Complexity

Time: O(n)

Space: O(1) (No extra space)